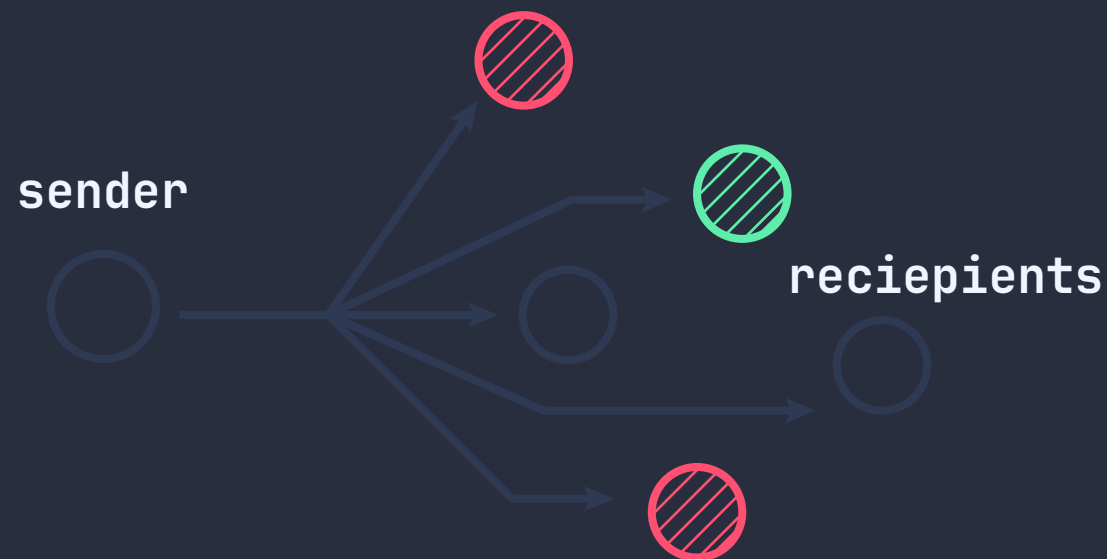


ROUTING SCHEMES

broadcast



Uses a one-to-all association where a single sender transmits a packet to all nodes on the network segment. The network replicates the packet as needed to reach every device within the subnet.

Often used in local networks for things like ARP requests or DHCP discovery.

unicast



Uses a one-to-one association between a sender and a single specific destination. Each packet is sent directly to one unique receiver identified by its address.

Used in everyday tasks like visiting a website or sending an email to one person.

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anycast



Uses a one-to-one-of-many association. A sender sends a packet to one receiver chosen from a group that shares the same address. The network delivers it to the nearest or most optimal node based on routing cost or distance.

Used in DNS and CDN services to improve speed and reliability by reaching the nearest server.

multicast



Uses a one-to-many-of-many or many-to-many-of-many association. A sender transmits a single packet to multiple selected receivers that have joined a specific multicast group. Only interested nodes receive the message.

Useful in streaming (e.g. IPTV) where only interested devices receive the data. Also used in routing protocols like: OSPF (uses 224.0.0.5 for AllSPFRouters, 224.0.0.6 for AllDRouters) EIGRP (uses 224.0.0.10 for all EIGRP-speaking routers)